Comparative Evaluation of Models for CO_2 Adsorption Prediction

Table of Model Performance

Model	R^2 Score	MAE	MSE	RMSE
Gradient Boosting	0.9304	0.1371	0.0276	0.1662
Random Forest	0.8048	0.2457	0.0775	0.2784
XGBoost	0.2657	0.5094	0.2915	0.5399
Decision Tree	0.2455	0.5150	0.2995	0.5472
AdaBoost	-0.1757	0.6750	0.4667	0.6831
MLP Regressor	-0.6989	0.8105	0.6743	0.8212
Support Vector	-0.7496	0.7315	0.6944	0.8333
RBF-like MLP	-1.1863	0.9302	0.8678	0.9315
Linear Regression	-1.5768	0.9269	1.0227	1.0113
Kernel Ridge	-6.7252	1.5132	3.0661	1.7510

Summary

Among the ten regression models tested to predict CO_2 adsorption capacity, the **Gradient Boosting Regressor** emerged as the top performer with an R^2 score of **0.93** and the lowest RMSE of **0.1662**, indicating both high accuracy and low prediction error. The **Random Forest Regressor** also performed strongly with an R^2 of **0.80**.

Ensemble tree-based models (Gradient Boosting, Random Forest, and XGBoost) consistently outperformed linear, neural, and kernel-based models. Simpler methods like **Linear Regression** and **Kernel Ridge Regression** struggled significantly with negative R^2 scores and high errors, suggesting they were unable to capture the underlying nonlinear patterns in the data.

Neural models (MLP and RBF-like variants) and SVR showed moderate performance but did not match the accuracy of tree-based methods. These results emphasize that, for this dataset, **boosting techniques** offer superior predictive performance due to their ability to model complex feature interactions in small datasets.